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Piaget’s Theory and Stages of Cognitive Development

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Key Takeaways

- Jean Piaget is famous for his theories regarding changes in cognitive development that occur as we move from infancy to adulthood.
- Cognitive development results from the interplay between innate capabilities (nature) and environmental influences (nurture).
- Children progress through four distinct stages, each representing varying cognitive abilities and world comprehension: the sensorimotor stage (birth to 2 years), the preoperational stage (2 to 7 years), the concrete operational stage (7 to 11 years), and the formal operational stage (11 years and beyond).
- A child’s cognitive development is not just about acquiring knowledge, the child has to develop or construct a mental model of the world, which is referred to as a schema.
- Piaget emphasized the role of active exploration and interaction with the environment in shaping cognitive development, highlighting the importance of assimilation and accommodation in constructing mental schemas.

Stages of Development

> Jean Piaget’s theory of cognitive development suggests that children move through four different stages of intellectual development which reflect the increasing sophistication of children’s thought

What is this?

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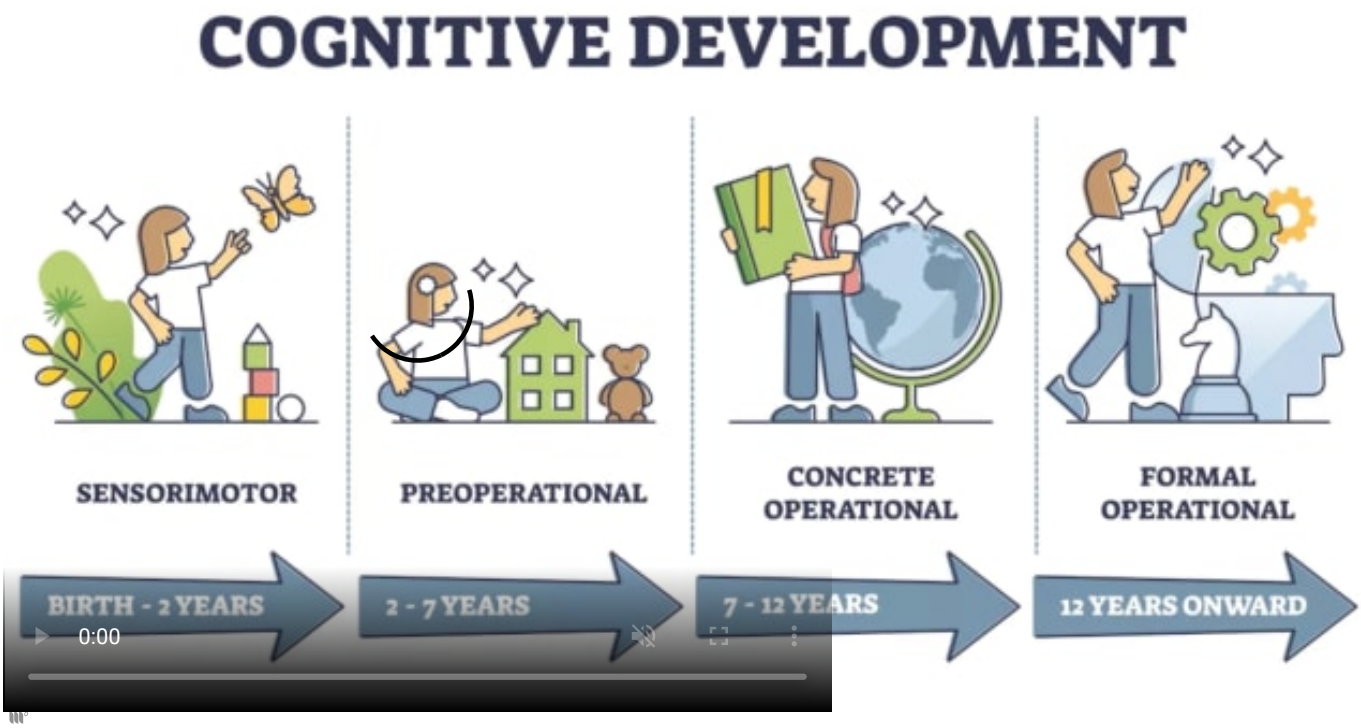
Each child goes through the stages in the same order (but not all at the same rate), and child development is determined by biological maturation and interaction with the environment.

Video Summary: Jean Piaget's Stages of Cognitive Development

Piaget Stages of Development



What is this?

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At each stage of development, the child's thinking is qualitatively different from the other stages, that is, each stage involves a different type of intelligence.

Stage	Age	Goal
Sensorimotor	Birth to 18-24 months	Object permanence
Preoperational	2 to 7 years old	Symbolic thought
Concrete operational	Ages 7 to 11 years	Logical thought
Formal operational	Adolescence to adulthood	Scientific reasoning

Piaget's 4 Stages of Cognitive Development

Although no stage can be missed out, there are individual differences in the rate at which children progress through stages, and some individuals may never attain the later stages.

Piaget did not claim that a particular stage was reached at a certain age – although descriptions of the stages often include an indication of the age at which the average child would reach each stage.

The Sensorimotor Stage

Ages: Birth to 2 Years

The first stage is the sensorimotor stage, during which the infant focuses on physical sensations and learning to coordinate its body.

Major Characteristics and Developmental Changes:

- The infant learns about the world through their senses and through their actions (moving around and exploring their environment).
- During the sensorimotor stage, a range of cognitive abilities develop. These include: object permanence; self-recognition (the child realizes that other people are separate from them); deferred imitation; and representational play.
- They relate to the emergence of the general symbolic function, which is the capacity to represent the world mentally
- At about 8 months, the infant will understand the permanence of objects and that they will still exist even if they can't see them and the infant will search for them when they disappear.

What is this?
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During the beginning of this stage, the infant lives in the present. It does not yet have a mental picture of the world stored in its memory therefore it does not have a sense of object permanence.

If it cannot see something, then it does not exist. This is why you can hide a toy from an infant, while it watches, but it will not search for the object once it has gone out of sight.

The main achievement during this stage is object permanence – knowing that an object still exists, even if it is hidden. It requires the ability to form a mental representation (i.e., a schema) of the object.

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> Towards the end of this stage the general symbolic function begins to appear where children show in their play that they can use one object to stand for another. Language starts to appear because they realise that words can be used to represent objects and feelings.

The child begins to be able to store information that it knows about the world, recall it, and label it.

Individual Differences

- **Cultural Practices**: In some cultures, babies are carried on their mothers' backs throughout the day. This constant physical contact and varied stimuli can influence how a child perceives their environment and their sense of object permanence.
- **Gender Norms**: Toys assigned to babies can differ based on gender expectations. A boy might be given more cars or action figures, while a girl might receive dolls or kitchen sets. This can influence early interactions and sensory explorations.

Learn More: The Sensorimotor Stage of Cognitive Development

The Preoperational Stage

Ages: 2 – 7 Years

Piaget's second stage of intellectual development is the preoperational stage. It takes place between 2 and 7 years. At the beginning of this stage, the child does not use operations, so the thinking is influenced by the way things appear rather than logical reasoning.

A child cannot conserve which means that the child does not understand that quantity remains the same even if the appearance changes.

What is this?
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Furthermore, the child is egocentric; he assumes that other people see the world as he does. This has been shown in the three mountains study.

As the preoperational stage develops, egocentrism declines, and children begin to enjoy the participation of another child in their games, and let's pretend play becomes more important.

Toddlers often pretend to be people they are not (e.g. superheroes, policemen), and may play these roles with props that symbolize real-life objects. Children may also invent an imaginary playmate.

Major Characteristics and Developmental Changes:

- Toddlers and young children acquire the ability to internally represent the world through language and mental imagery.
- During this stage, young children can think about things symbolically. This is the ability to make one thing, such as a word or an object, stand for something other than itself.
- A child's thinking is dominated by how the world looks, not how the world is. It is not yet capable of logical (problem-solving) type of thought.
- Moreover, the child has difficulties with class inclusion; he can classify objects but cannot include objects in sub-sets, which involves classifying objects as belonging to two or more categories simultaneously.
- Infants at this stage also demonstrate animism. This is the tendency for the child to think that non-living objects (such as toys) have life and feelings like a person's.

What is this?

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By 2 years, children have made some progress toward detaching their thoughts from the physical world. However, have not yet developed logical (or "operational") thought characteristics of later stages.

Thinking is still intuitive (based on subjective judgments about situations) and egocentric (centered on the child's own view of the world).

Individual Differences

- **Cultural Storytelling**: Different cultures have unique stories, myths, and folklore. Children from diverse backgrounds might understand and interpret symbolic elements differently based on their cultural narratives.
- **Race & Representation**: A child's racial identity can influence how they engage in pretend play. For instance, a lack of diverse representation in media and toys might lead children of color to recreate scenarios that don't reflect their experiences or background.

Learn More: The Preoperational Stage of Cognitive Development

The Concrete Operational Stage

Ages: 7 – 11 Years

By the beginning of the concrete operational stage, the child can use operations (a set of logical rules) so they can conserve quantities, realize that people see the world in a different way (decentring), and demonstrate improvement in inclusion tasks. Children still have difficulties with abstract thinking.

Major Characteristics and Developmental Changes:

- During this stage, children begin to think logically about concrete events.
- Children begin to understand the concept of conservation; understanding that, although things may change in appearance, certain properties remain the same.
- During this stage, children can mentally reverse things (e.g., picture a ball of plasticine returning to its original shape).
- During this stage, children also become less egocentric and begin to think about how other people might think and feel.

What is this?
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The stage is called concrete because children can think logically much more successfully if they can manipulate real (concrete) materials or pictures of them.

Piaget considered the concrete stage a major turning point in the child's cognitive development because it marks the beginning of logical or operational thought. This means the child can work things out internally in their head (rather than physically try things out in the real world).

What is this?
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Children can conserve number (age 6), mass (age 7), and weight (age 9). Conservation is the understanding that something stays the same in quantity even though its appearance changes.

But operational thought is only effective here if the child is asked to reason about materials that are physically present. Children at this stage will tend to make mistakes or be overwhelmed when asked to reason about abstract or hypothetical problems.

Individual Differences

- **Cultural Context in Conservation Tasks**: In a society where resources are scarce, children might demonstrate conservation skills earlier due to the cultural emphasis on preserving and reusing materials.
- **Gender & Learning**: Stereotypes about gender abilities, like “boys are better at math,” can influence how children approach logical problems or classify objects based on perceived gender norms.

Learn More: The Concrete Operational Stage of Development

The Formal Operational Stage

Ages: 12 and Over

The formal operational period begins at about age 11. As adolescents enter this stage, they gain the ability to think in an abstract manner, the ability to combine and classify items in a more sophisticated way, and the capacity for higher-order reasoning.

Adolescents can think systematically and reason about what might be as well as what is (not everyone achieves this stage). This allows them to understand politics, ethics, and science fiction, as well as to engage in scientific reasoning.

Adolescents can deal with abstract ideas: e.g. they can understand division and fractions without having to actually divide things up, and solve hypothetical (imaginary) problems.

Major Characteristics and Developmental Changes:

- Concrete operations are carried out on things whereas formal operations are carried out on ideas. Formal operational thought is entirely freed from physical and perceptual constraints.
- During this stage, adolescents can deal with abstract ideas (e.g. no longer needing to think about slicing up cakes or sharing sweets to understand division and fractions).
- They can follow the form of an argument without having to think in terms of specific examples.
- Adolescents can deal with hypothetical problems with many possible solutions. E.g. if asked ‘What would happen if money were abolished in one hour’s time? they could speculate about many possible consequences.

What is this?
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From about 12 years children can follow the form of a logical argument without reference to its content. During this time, people develop the ability to think about abstract concepts, and logically test hypotheses.

This stage sees the emergence of scientific thinking, formulating abstract theories and hypotheses when faced with a problem.

Individual Differences

- **Culture & Abstract Thinking**: Cultures emphasize different kinds of logical or abstract thinking. For example, in societies with a strong oral tradition, the ability to hold complex narratives might develop prominently.
- **Gender & Ethics**: Discussions about morality and ethics can be influenced by gender norms. For instance, in some cultures, girls might be encouraged to prioritize community harmony, while boys might be encouraged to prioritize individual rights.

What is this?
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Learn More: The Formal Operational Stage of Development

Piaget's Theory

- Piaget's theory places a strong emphasis on the active role that children play in their own cognitive development.
- According to Piaget, children are not passive recipients of information; instead, they actively explore and interact with their surroundings.
- This active engagement with the environment is crucial because it allows them to gradually build their understanding of the world.

1. How Piaget Developed the Theory

Piaget was employed at the Binet Institute in the 1920s, where his job was to develop French versions of questions on English intelligence tests. He became intrigued with the reasons children gave for their wrong answers to the questions that required logical thinking.

He believed that these incorrect answers revealed important differences between the thinking of adults and children.

Piaget branched out on his own with a new set of assumptions about children's intelligence:

- Children's intelligence differs from an adult's in quality rather than in quantity. This means that children reason (think) differently from adults and see the world in different ways.
- Children actively build up their knowledge about the world. They are not passive creatures waiting for someone to fill their heads with knowledge.
- The best way to understand children's reasoning is to see things from their point of view.

What is this?
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Piaget did not want to measure how well children could count, spell or solve problems as a way of grading their I.Q. What he was more interested in was the way in which fundamental concepts like the very idea of number, time, quantity, causality, justice, and so on emerged.

Piaget studied children from infancy to adolescence using naturalistic observation of his own three babies and sometimes controlled observation too. From these, he wrote diary descriptions charting their development.

He also used clinical interviews and observations of older children who were able to understand questions and hold conversations.

2. Piaget's Theory Differs From Others In Several Ways:

Piaget's (1936, 1950) theory of cognitive development explains how a child constructs a mental model of the world. He disagreed with the idea that intelligence was a fixed trait, and regarded cognitive development as a process that occurs due to biological maturation and interaction with the environment.

Children's ability to understand, think about, and solve problems in the world develops in a stop-start, discontinuous manner (rather than gradual changes over time).

- It is concerned with children, rather than all learners.
- It focuses on development, rather than learning per se, so it does not address learning of information or specific behaviors.
- It proposes discrete stages of development, marked by qualitative differences, rather than a gradual increase in number and complexity of behaviors, concepts, ideas, etc.

What is this?

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The goal of the theory is to explain the mechanisms and processes by which the infant, and then the child, develops into an individual who can reason and think using hypotheses.

To Piaget, cognitive development was a progressive reorganization of mental processes as a result of biological maturation and environmental experience.

Children construct an understanding of the world around them, then experience discrepancies between what they already know and what they discover in their environment.

3. Schemas

Piaget claimed that knowledge cannot simply emerge from sensory experience; some initial structure is necessary to make sense of the world.

> According to Piaget, children are born with a very basic mental structure (genetically inherited and evolved) on which all subsequent learning and knowledge are based.

Schemas are the basic building blocks of such cognitive models, and enable us to form a mental representation of the world.

Piaget (1952, p. 7) defined a schema as: "a cohesive, repeatable action sequence possessing component actions that are tightly interconnected and governed by a core meaning."

In more simple terms, Piaget called the schema the basic building block of intelligent behavior – a way of organizing knowledge. Indeed, it is useful to think of schemas as “units” of knowledge, each relating to one aspect of the world, including objects, actions, and abstract (i.e., theoretical) concepts.

Wadsworth (2004) suggests that schemata (the plural of schema) be thought of as “index cards” filed in the brain, each one telling an individual how to react to incoming stimuli or information.

When Piaget talked about the development of a person’s mental processes, he was referring to increases in the number and complexity of the schemata that a person had learned.

What is this?
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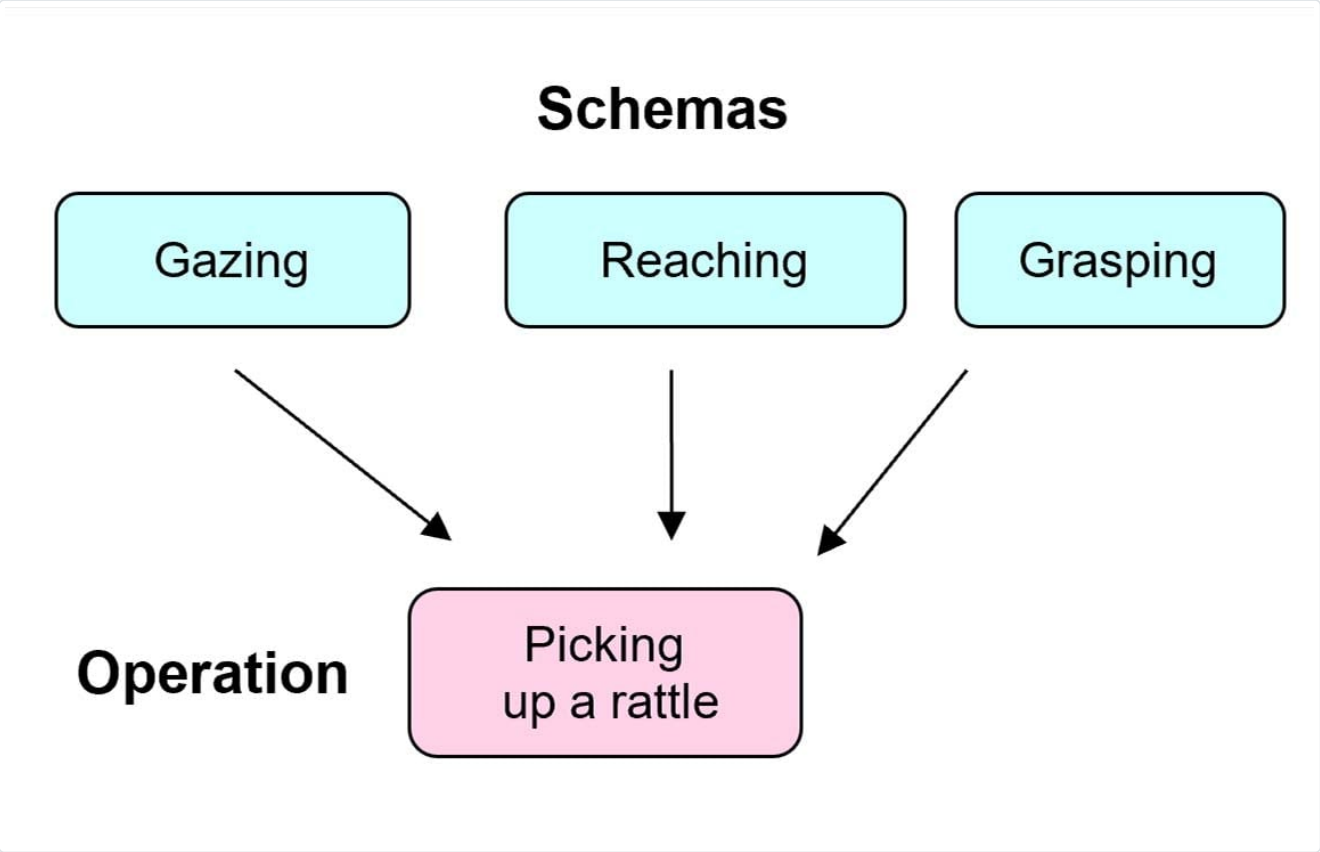
> When a child’s existing schemas are capable of explaining what it can perceive around it, it is said to be in a state of equilibrium, i.e., a state of cognitive (i.e., mental) balance.

Operations are more sophisticated mental structures which allow us to combine schemas in a logical (reasonable) way.

As children grow they can carry out more complex operations and begin to imagine hypothetical (imaginary) situations.

Apart from the schemas we are born with schemas and operations are learned through interaction with other people and the environment.



What is this?
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Piaget emphasized the importance of schemas in cognitive development and described how they were developed or acquired.

A schema can be defined as a set of linked mental representations of the world, which we use both to understand and to respond to situations. The assumption is that we store these mental representations and apply them when needed.

Examples of Schemas

A person might have a schema about buying a meal in a restaurant. The schema is a stored form of the pattern of behavior which includes looking at a menu, ordering food, eating it and paying the bill.

This is an example of a schema called a “script.” Whenever they are in a restaurant, they retrieve this schema from memory and apply it to the situation.

The schemas Piaget described tend to be simpler than this – especially those used by infants. He described how – as a child gets older – his or her schemas become more numerous and elaborate.

Piaget believed that newborn babies have a small number of innate schemas – even before they have had many opportunities to experience the world. These neonatal schemas are the cognitive structures underlying innate reflexes. These reflexes are genetically programmed into us.

For example, babies have a sucking reflex, which is triggered by something touching the baby’s lips. A baby will suck a nipple, a comforter (dummy), or a person’s finger. Piaget, therefore, assumed that the baby has a “sucking schema.”

Similarly, the grasping reflex which is elicited when something touches the palm of a baby’s hand, or the rooting reflex, in which a baby will turn its head towards something which touches its cheek, are innate schemas. Shaking a rattle would be the combination of two schemas, grasping and shaking.

4. The Process of Adaptation

Piaget also believed that a child developed as a result of two different influences: maturation, and interaction with the environment. The child develops mental structures (schemata) which enables him to solve problems in the environment.

What is this?

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Adaptation is the process by which the child changes its mental models of the world to match more closely how the world actually is.

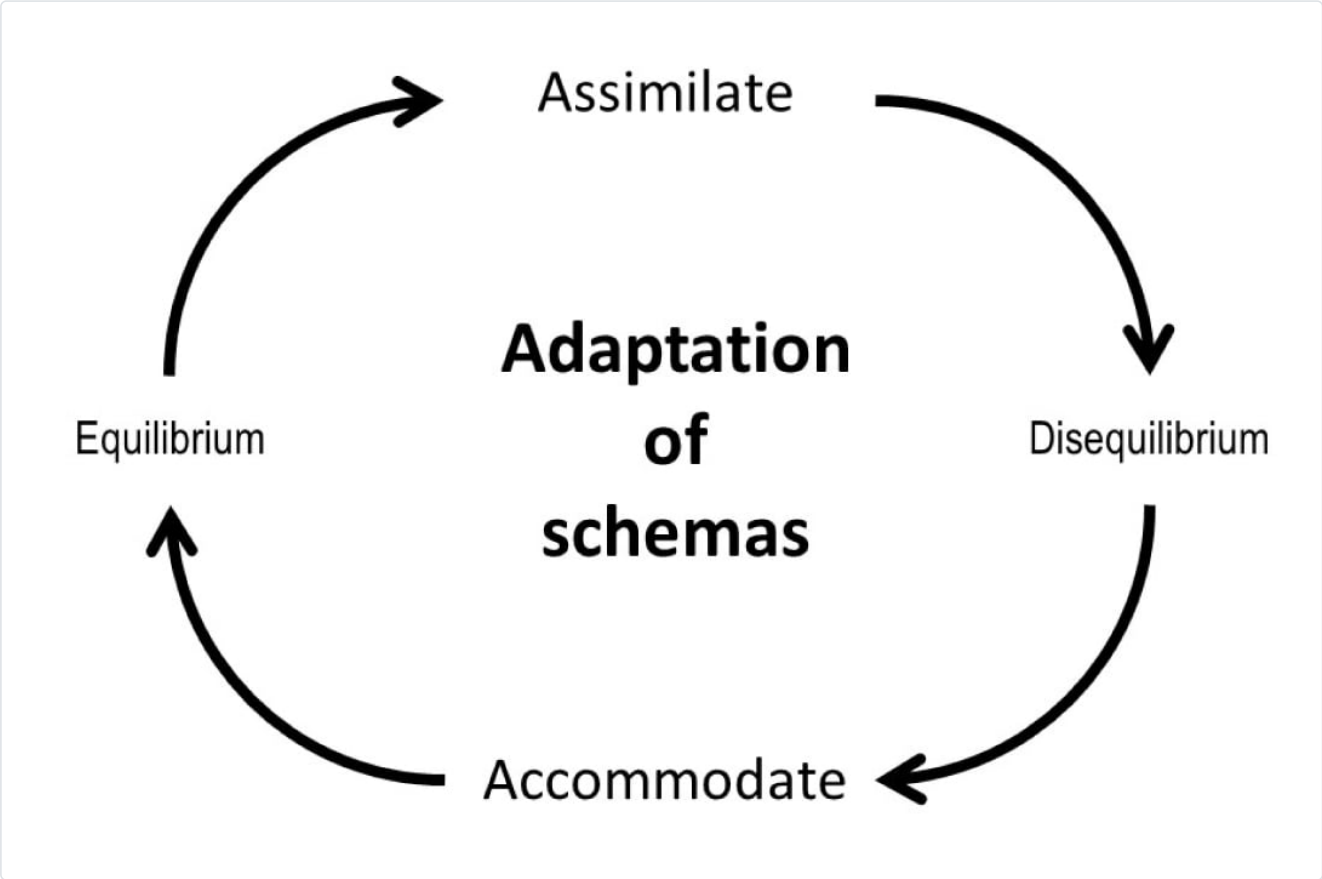
Adaptation is brought about by the processes of assimilation (solving new experiences using existing schemata) and accommodation (changing existing schemata in order to solve new experiences).

The importance of this viewpoint is that the child is seen as an active participant in its own development rather than a passive recipient of either biological influences (maturation) or environmental stimulation.

When our existing schemas can explain what we perceive around us, we are in a state of **equilibration**. However, when we meet a new situation that we cannot explain it creates disequilibrium, this is an unpleasant sensation which we try to escape, and this gives us the motivation to learn.

According to Piaget, reorganization to higher levels of thinking is not accomplished easily. The child must “rethink” his or her view of the world. An important step in the process is the experience of cognitive conflict.

In other words, the child becomes aware that he or she holds two contradictory views about a situation and they both cannot be true. This step is referred to as **disequilibrium**.



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> Jean Piaget (1952; see also Wadsworth, 2004) viewed intellectual growth as a process of **adaptation** (adjustment) to the world. This happens through assimilation, accommodation, and equilibration.

To get back to a state of equilibration, we need to modify our existing schemas to learn and adapt to the new situation.

This is done through the processes of accommodation and assimilation. This is how our schemas evolve and become more sophisticated. The processes of assimilation and accommodation are continuous and interactive.

5. Assimilation

Piaget defined assimilation as the cognitive process of fitting new information into existing cognitive schemas, perceptions, and understanding. Overall beliefs and understanding of the world do not change as a result of the new information.

Assimilation occurs when the new experience is not very different from previous experiences of a particular object or situation we assimilate the new situation by adding information to a previous schema.

This means that when you are faced with new information, you make sense of this information by referring to information you already have (information processed and learned previously) and trying to fit the new information into the information you already have.

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Examples

1. Imagine a young child who has only ever seen small, domesticated dogs. When the child sees a cat for the first time, they might refer to it as a “dog” because it has four legs, fur, and a tail – features that fit their existing schema of a dog.
2. A person who has always believed that all birds can fly might label penguins as birds that can fly. This is because their existing schema or understanding of birds includes the ability to fly.
3. A 2-year-old child sees a man who is bald on top of his head and has long frizzy hair on the sides. To his father’s horror, the toddler shouts “Clown, clown” (Siegler et al., 2003).
4. If a baby learns to pick up a rattle he or she will then use the same schema (grasping) to pick up other objects.

6. Accommodation

Accommodation: when the new experience is very different from what we have encountered before we need to change our schemas in a very radical way or create a whole new schema.

Psychologist Jean Piaget defined accommodation as the cognitive process of revising existing cognitive schemas, perceptions, and understanding so that new information can be incorporated.

This happens when the existing schema (knowledge) does not work, and needs to be changed to deal with a new object or situation.

In order to make sense of some new information, you actually adjust information you already have (schemas you already have, etc.) to make room for this new information.

Examples

1. A baby tries to use the same schema for grasping to pick up a very small object. It doesn’t work. The baby then changes the schema by now using the forefinger and thumb to pick up the object.
2. A child may have a schema for birds (feathers, flying, etc.) and then they see a plane, which also flies, but would not fit into their bird schema.
3. In the “clown” incident, the boy’s father explained to his son that the man was not a clown and that even though his hair was like a clown’s, he wasn’t wearing a funny costume and wasn’t doing silly things to make people laugh. With this new knowledge, the boy was able to change his schema of “clown” and make this idea fit better to a standard concept of “clown”.
4. A person who grew up thinking all snakes are dangerous might move to an area where garden snakes are common and harmless. Over time, after observing and learning, they might accommodate their previous belief to understand that not all snakes are harmful.

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7. Equilibration

Piaget believed that all human thought seeks order and is uncomfortable with contradictions and inconsistencies in knowledge structures. In other words, we seek “equilibrium” in our cognitive structures.

Equilibrium occurs when a child’s schemas can deal with most new information through assimilation. However, an unpleasant state of disequilibrium occurs when new information cannot be fitted into existing schemas (assimilation).

Piaget believed that cognitive development did not progress at a steady rate, but rather in leaps and bounds. Equilibration is the force which drives the learning process as we do not like to be frustrated and will seek to restore balance by mastering the new challenge (accommodation).

Once the new information is acquired the process of assimilation with the new schema will continue until the next time we need to make an adjustment to it.

Equilibration is a regulatory process that maintains a balance between assimilation and accommodation to facilitate cognitive growth. Think of it this way: We can’t merely assimilate all the time; if we did, we would never learn any new concepts or principles.

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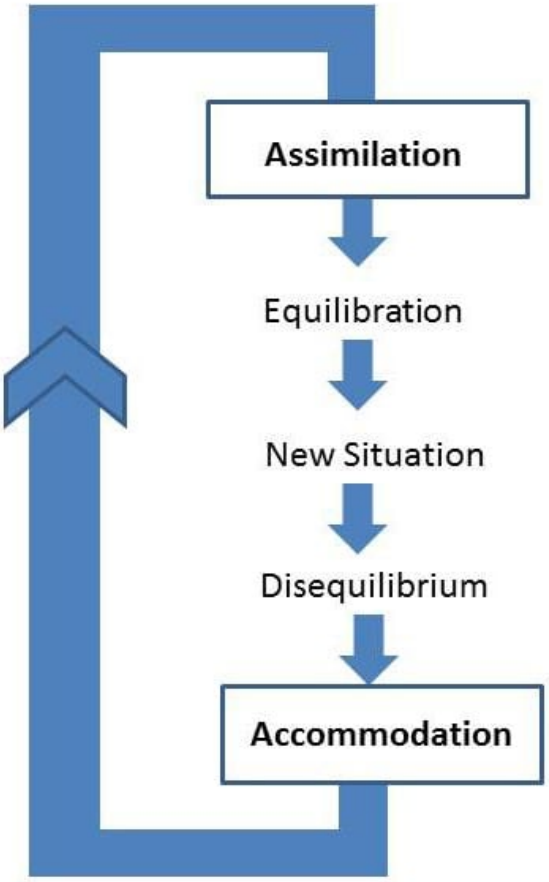
Everything new we encountered would just get put in the same few “slots” we already had. Neither can we accommodate all the time; if we did, everything we encountered would seem new; there would be no recurring regularities in our world. We’d be exhausted by the mental effort!



Applications to Education

Think of old black and white films that you've seen in which children sat in rows at desks, with ink wells, would learn by rote, all chanting in unison in response to questions set by an authoritarian old biddy like Matilda!

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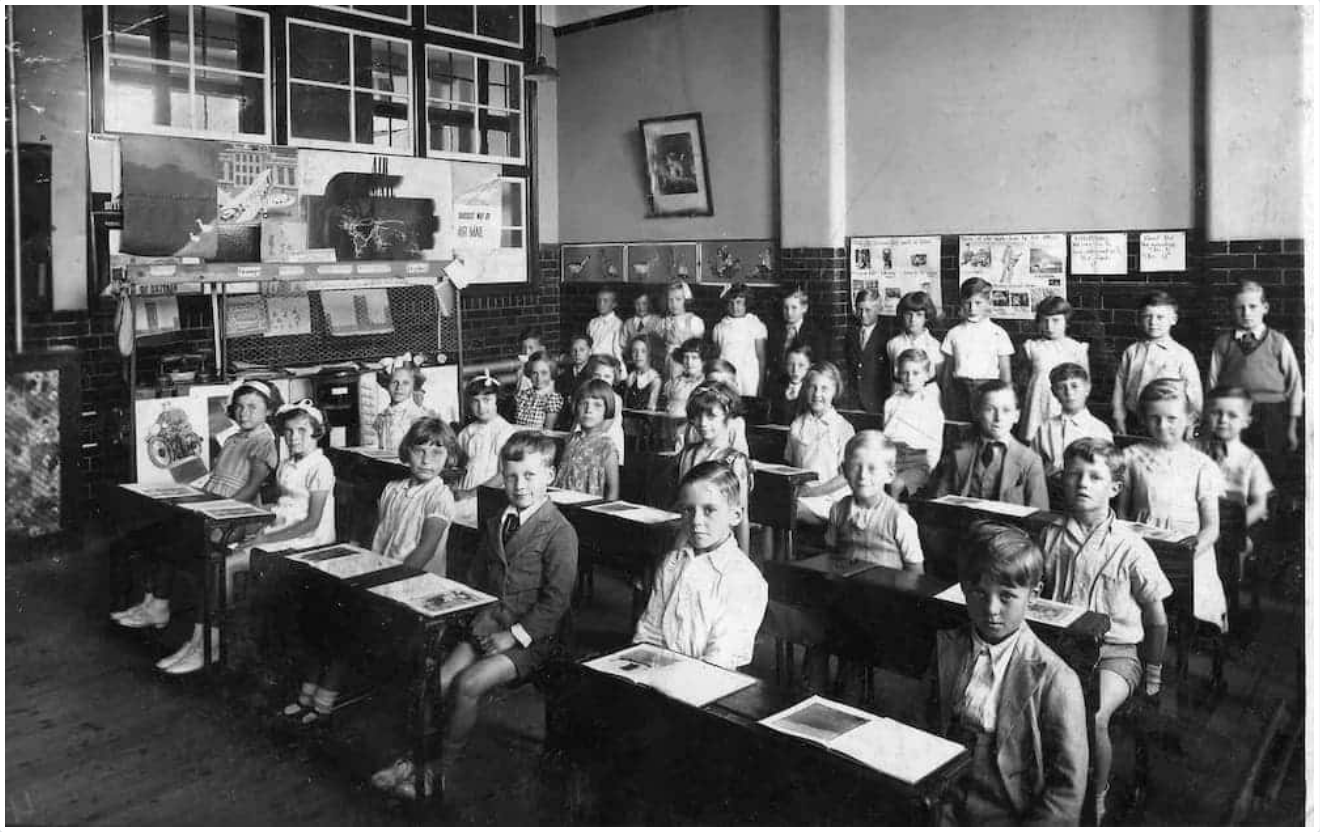
Children who were unable to keep up were seen as slacking and would be punished by variations on the theme of corporal punishment. Yes, it really did happen and in some parts of the world still does today.

Piaget is partly responsible for the change that occurred in the 1960s and for your relatively pleasurable and pain-free school days!



> "Children should be able to do their own experimenting and their own research. Teachers, of course, can guide them by providing appropriate materials, but the essential thing is that in order for a child to understand something, he must construct it himself, he must re-invent it. Every time we teach a child something, we keep him from inventing it himself. On the other hand that which we allow him to discover by himself will remain with him visibly".
>
> *Piaget (1972, p. 27)*

What is this?

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Plowden Report

Piaget (1952) did not explicitly relate his theory to education, although later researchers have explained how features of Piaget's theory can be applied to teaching and learning.

Piaget has been extremely influential in developing educational policy and teaching practice. For example, a review of primary education by the UK government in 1966 was based strongly on Piaget's theory. The result of this review led to the publication of the Plowden Report (1967).

In the 1960s the Plowden Committee investigated the deficiencies in education and decided to incorporate many of Piaget's ideas into its final report published in 1967, even though Piaget's work was not really designed for education.

> The report makes three Piaget-associated recommendations:

1. Children should be given individual attention and it should be realized that they need to be treated differently.
2. Children should only be taught things that they are capable of learning
3. Children mature at different rates and the teacher needs to be aware of the stage of development of each child so teaching can be tailored to their individual needs.

"The report's recurring themes are individual learning, flexibility in the curriculum, the centrality of play in children's learning, the use of the environment, learning by discovery and the importance of the evaluation of children's progress – teachers should "not assume that only what is measurable is valuable."

Discovery learning – the idea that children learn best through doing and actively exploring – was seen as central to the transformation of the primary school curriculum.

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How to teach

Within the classroom learning should be student-centered and accomplished through active discovery learning. The role of the teacher is to facilitate learning, rather than direct tuition.

Because Piaget's theory is based upon biological maturation and stages, the notion of "readiness" is important. Readiness concerns when certain information or concepts should be taught.

According to Piaget's theory, children should not be taught certain concepts until they have reached the appropriate stage of cognitive development.

According to Piaget (1958), assimilation and accommodation require an active learner, not a passive one, because problem-solving skills cannot be taught, they must be discovered.

> Therefore, teachers should encourage the following within the classroom:

1. Educational programs should be designed to correspond to Piaget's stages of development. Children in the concrete operational stage should be given concrete means to learn new concepts e.g. tokens for counting.
2. Devising situations that present useful problems, and create disequilibrium in the child.
3. Focus on the process of learning, rather than the end product of it. Instead of checking if children have the right answer, the teacher should focus on the student's understanding and the processes they used to get to the answer.

4. Child-centered approach. Learning must be active (discovery learning).
   Children should be encouraged to discover for themselves and to interact with the material instead of being given ready-made knowledge.
5. Accepting that children develop at different rates so arrange activities for individual children or small groups rather than assume that all the children can cope with a particular activity.
6. Using active methods that require rediscovering or reconstructing “truths.”
7. Using collaborative, as well as individual activities (so children can learn from each other).
8. Evaluate the level of the child’s development so suitable tasks can be set.
9. Adapt lessons to suit the needs of the individual child (i.e. differentiated teaching).
10. Be aware of the child’s stage of development (testing).
11. Teach only when the child is ready. i.e. has the child reached the appropriate stage.
12. Providing support for the “spontaneous research” of the child.
13. Using collaborative, as well as individual activities.
14. Devising situations that present useful problems, and create disequilibrium in the child.
15. Educators may use Piaget’s stages to design age-appropriate assessment tools and strategies.

What is this?
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Classroom Activities

1. **Sensorimotor Stage (0-2 years):**

   Although most kids in this age range are not in a traditional classroom setting, they can still benefit from games that stimulate their senses and motor skills.

   - **Object Permanence Games**: Play peek-a-boo or hide toys under a blanket to help babies understand that objects still exist even when they can’t see them.
   - **Sensory Play**: Activities like water play, sand play, or playdough encourage exploration through touch.
   - **Imitation**: Children at this age love to imitate adults. Use imitation as a way to teach new skills.

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2. **Preoperational Stage (2-7 years):**

   - **Role Playing**: Set up pretend play areas where children can act out different scenarios, such as a kitchen, hospital, or market.
   - **Use of Symbols**: Encourage drawing, building, and using props to represent other things.
   - **Hands-on Activities**: Children should interact physically with their environment, so provide plenty of opportunities for hands-on learning.
   - **Egocentrism Activities**: Use exercises that highlight different perspectives. For instance, having two children sit across from each other with an object in between and asking them what the other sees.

3. **Concrete Operational Stage (7-11 years):**
   - **Classification Tasks**: Provide objects or pictures to group, based on various characteristics.
   - **Hands-on Experiments**: Introduce basic science experiments where they can observe cause and effect, like a simple volcano with baking soda and vinegar.
   - **Logical Games**: Board games, puzzles, and logic problems help develop their thinking skills.
   - **Conservation Tasks**: Use experiments to showcase that quantity doesn't change with alterations in shape, such as the classic liquid conservation task using different shaped glasses.

4. **Formal Operational Stage (11 years and older):**
   - **Hypothesis Testing**: Encourage students to make predictions and test them out.
   - **Abstract Thinking**: Introduce topics that require abstract reasoning, such as algebra or ethical dilemmas.
   - **Problem Solving**: Provide complex problems and have students work on solutions, integrating various subjects and concepts.
   - **Debate and Discussion**: Encourage group discussions and debates on abstract topics, highlighting the importance of logic and evidence.
   - **Feedback and Questioning**: Use open-ended questions to challenge students and promote higher-order thinking. For instance, rather than asking, "Is this the right answer?", ask, "How did you arrive at this conclusion?"

What is this?
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Individual Differences

While Piaget's stages offer a foundational framework, they are not universally experienced in the same way by all children.

Social identities play a critical role in shaping cognitive development, necessitating a more nuanced and culturally responsive approach to understanding child development.

> Piaget's stages may manifest differently based on social identities like race, gender, and culture:

- **Race & Teacher Interactions**: A child's race can influence teacher expectations and interactions. For example, racial biases can lead to children of color being perceived as less capable or more disruptive, influencing their cognitive challenges and supports.
- **Racial and Cultural Stereotypes**: These can affect a child's self-perception and self-efficacy. For instance, stereotypes about which racial or cultural groups are "better" at certain subjects can influence a child's self-confidence and, subsequently, their engagement in that subject.
- **Gender & Peer Interactions**: Children learn gender roles from their peers. Boys might be mocked for playing "girl games," and girls might be excluded from certain activities, influencing their cognitive engagements.
- **Language**: Multilingual children might navigate the stages differently, especially if their home language differs from their school language. The way concepts are framed in different languages can influence cognitive processing. Cultural idioms and metaphors can shape a child's understanding of concepts and their ability to use symbolic representation, especially in the pre-operational stage.

What is this?

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Curriculum Development

According to Piaget, children's cognitive development is determined by a process of maturation which cannot be altered by tuition so education should be stage-specific.

For example, a child in the concrete operational stage should not be taught abstract concepts and should be given concrete aid such as tokens to count with.

According to Piaget children learn through the process of accommodation and assimilation so the role of the teacher should be to provide opportunities for these processes to occur such as new material and experiences that challenge the children's existing schemas.

Furthermore, according to this theory, children should be encouraged to discover for themselves and to interact with the material instead of being given ready-made knowledge.

Curricula need to be developed that take into account the age and stage of thinking of the child. For example there is no point in teaching abstract concepts such as algebra or atomic structure to children in primary school.

Curricula also need to be sufficiently flexible to allow for variations in the ability of different students of the same age. In Britain, the National Curriculum and Key Stages broadly reflect the stages that Piaget laid down.

For example, egocentrism dominates a child's thinking in the sensorimotor and preoperational stages. Piaget would therefore predict that using group activities would not be appropriate since children are not capable of understanding the views of others.

However, Smith et al. (1998), point out that some children develop earlier than Piaget predicted and that by using group work children can learn to appreciate the views of others in preparation for the concrete operational stage.

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The national curriculum emphasizes the need to use concrete examples in the primary classroom.

Shayer (1997), reported that abstract thought was necessary for success in secondary school (and co-developed the CASE system of teaching science). Recently the National curriculum has been updated to encourage the teaching of some abstract concepts towards the end of primary education, in preparation for secondary courses. (DfEE, 1999).

Child-centered teaching is regarded by some as a child of the 'liberal sixties.' In the 1980s the Thatcher government introduced the National Curriculum in an attempt to move away from this and bring more central government control into the teaching of children.

So, although the British National Curriculum in some ways supports the work of Piaget, (in that it dictates the order of teaching), it can also be seen as prescriptive to the point where it counters Piaget's child-oriented approach.

However, it does still allow for flexibility in teaching methods, allowing teachers to tailor lessons to the needs of their students.

Social Media (Digital Learning)

Jean Piaget could not have anticipated the expansive digital age we now live in.

Today, knowledge dissemination and creation are democratized by the Internet, with platforms like blogs, wikis, and social media allowing for vast collaboration and shared knowledge. This development has prompted a reimagining of the future of education.

Classrooms, traditionally seen as primary sites of learning, are being overshadowed by the rise of mobile technologies and platforms like MOOCs (Passey, 2013).

The millennial generation, defined as the first to grow up with cable TV, the internet, and cell phones, relies heavily on technology.

They view it as an integral part of their identity, with most using it extensively in their daily lives, from keeping in touch with loved ones to consuming news and entertainment (Nielsen, 2014).

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Social media platforms offer a dynamic environment conducive to Piaget's principles. These platforms allow for interactions that nurture knowledge evolution through cognitive processes like assimilation and accommodation.

They emphasize communal interaction and shared activity, fostering both cognitive and socio-cultural constructivism. This shared activity promotes understanding and exploration beyond individual perspectives, enhancing social-emotional learning (Gehlbach, 2010).

A standout advantage of social media in an educational context is its capacity to extend beyond traditional classroom confines. As the material indicates, these platforms can foster more inclusive learning, bridging diverse learner groups.

This inclusivity can equalize learning opportunities, potentially diminishing biases based on factors like race or socio-economic status, resonating with Kegan's (1982) concept of "recruitability."

However, there are challenges. While the potential of social media in learning is vast, its practical application necessitates intention and guidance. Cuban, Kirkpatrick, and Peck (2001) note that certain educators and students are hesitant about integrating social media into educational contexts.

This hesitancy can stem from technological complexities or potential distractions. Yet, when harnessed effectively, social media can provide a rich environment for collaborative learning and interpersonal development, fostering a deeper understanding of content.

In essence, the rise of social media aligns seamlessly with constructivist philosophies. Social media platforms act as tools for everyday cognition, merging daily social interactions with the academic world, and providing avenues for diverse, interactive, and engaging learning experiences.

Applications to Parenting

Parents can use Piaget's stages to have realistic developmental expectations of their children's behavior and cognitive capabilities.

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For instance, understanding that a toddler is in the pre-operational stage can help parents be patient when the child is egocentric.

Play Activities

Recognizing the importance of play in cognitive development, many parents provide toys and games suited for their child's developmental stage.

Parents can offer activities that are slightly beyond their child's current abilities, leveraging Vygotsky's concept of the "Zone of Proximal Development," which complements Piaget's ideas.

1. **Sensorimotor Stage (0-2 years):**
   - **Peek-a-boo**: Helps with object permanence.
   - **Texture Touch**: Provide different textured materials (soft, rough, bumpy, smooth) for babies to touch and feel.
   - **Sound Bottles**: Fill small bottles with different items like rice, beans, bells, and have children shake and listen to the different sounds.
2. **Preoperational Stage (2-7 years):**
   - **Memory Games**: Using cards with pictures, place them face down, and ask students to find matching pairs.
   - **Role Playing and Pretend Play**: Let children act out roles or stories that enhance symbolic thinking. Encourage symbolic play with dress-up clothes, playsets, or toy cash registers. Provide prompts or scenarios to extend their imagination.
   - **Story Sequencing**: Give children cards with parts of a story and have them arranged in the correct order.
3. **Concrete Operational Stage (7-11 years):**
   - **Number Line Jumps**: Create a number line on the floor with tape. Ask students to jump to the correct answer for math problems.
   - **Classification Games**: Provide a mix of objects and ask students to classify them based on different criteria (e.g., color, size, shape).
   - **Logical Puzzle Games**: Games that involve problem-solving using logic, such as simple Sudoku puzzles or logic grid puzzles.
4. **Formal Operational Stage (11 years and older):**
   - **Debate and Discussion**: Provide a topic and let students debate on pros and cons. This promotes abstract thinking and logical reasoning.
   - **Hypothesis Testing Games**: Present a scenario and have students come up with hypotheses and ways to test them.
   - **Strategy Board Games**: Games like chess, checkers, or Settlers of Catan can help in developing strategic and forward-thinking skills.

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Critical Evaluation

Support

> The influence of Piaget's ideas on developmental psychology has been enormous. He changed how people viewed the child's world and their methods of studying children.

He was an inspiration to many who came after and took up his ideas. Piaget's ideas have generated a huge amount of research which has increased our understanding of cognitive development.

- Piaget (1936) was one of the first psychologists to make a systematic study of cognitive development. His contributions include a stage theory of child cognitive development, detailed observational studies of cognition in children, and a series of simple but ingenious tests to reveal different cognitive abilities.
- His ideas have been of practical use in understanding and communicating with children, particularly in the field of education (re: Discovery Learning). Piaget's theory has been applied across education.
- According to Piaget's theory, educational programs should be designed to correspond to the stages of development.

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Criticisms

> Are the stages real? Vygotsky and Bruner would rather not talk about stages at all, preferring to see development as a continuous process. Others have queried the age ranges of the stages. Some studies have shown that progress to the formal operational stage is not guaranteed.

For example, Keating (1979) reported that 40-60% of college students fail at formal operation tasks, and Dasen (1994) states that only one-third of adults ever reach the formal operational stage.

The fact that the formal operational stage is not reached in all cultures and not all individuals within cultures suggests that it might not be biologically based.

- According to Piaget, the rate of cognitive development cannot be accelerated as it is based on biological processes however, direct tuition can speed up the development which suggests that it is not entirely based on biological factors.
- Because Piaget concentrated on the universal stages of cognitive development and biological maturation, he failed to consider the effect that the social setting and culture may have on cognitive development.

Cross-cultural studies show that the stages of development (except the formal operational stage) occur in the same order in all cultures suggesting that cognitive development is a product of a biological process of maturation.

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However, the age at which the stages are reached varies between cultures and individuals which suggests that social and cultural factors and individual differences influence cognitive development.

Dasen (1994) cites studies he conducted in remote parts of the central Australian desert with 8-14-year-old Indigenous Australians. He gave them conservation of liquid tasks and spatial awareness tasks. He found that the ability to conserve came later in the Aboriginal children, between ages of 10 and 13 (as opposed to between 5 and 7, with Piaget's Swiss sample).

However, he found that spatial awareness abilities developed earlier amongst the Aboriginal children than the Swiss children. Such a study demonstrates cognitive development is not purely dependent on maturation but on cultural factors too – spatial awareness is crucial for nomadic groups of people.

Vygotsky, a contemporary of Piaget, argued that social interaction is crucial for cognitive development. According to Vygotsky the child's learning always occurs in a social context in cooperation with someone more skillful (MKO). This social interaction provides language opportunities and Vygotsky considered language the foundation of thought.

> Piaget's methods (observation and clinical interviews) are more open to biased interpretation than other methods. Piaget made careful, detailed naturalistic observations of children, and from these, he wrote diary descriptions charting their development. He also used clinical interviews and observations of older children who were able to understand questions and hold conversations.

Because Piaget conducted the observations alone the data collected are based on his own subjective interpretation of events. It would have been more reliable if Piaget conducted the observations with another researcher and compared the results afterward to check if they are similar (i.e., have inter-rater reliability).

Although clinical interviews allow the researcher to explore data in more depth, the interpretation of the interviewer may be biased.

For example, children may not understand the question/s, they have short attention spans, they cannot express themselves very well, and may be trying to please the experimenter. Such methods meant that Piaget may have formed inaccurate conclusions.

> As several studies have shown Piaget underestimated the abilities of children because his tests were sometimes confusing or difficult to understand (e.g., Hughes, 1975).

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Piaget failed to distinguish between competence (what a child is capable of doing) and performance (what a child can show when given a particular task). When tasks were altered, performance (and therefore competence) was affected. Therefore, Piaget might have underestimated children's cognitive abilities.

For example, a child might have object permanence (competence) but still not be able to search for objects (performance). When Piaget hid objects from babies he found that it wasn't till after nine months that they looked for it.

However, Piaget relied on manual search methods – whether the child was looking for the object or not.

Later, researchers such as Baillargeon and Devos (1991) reported that infants as young as four months looked longer at a moving carrot that didn't do what it expected, suggesting they had some sense of permanence, otherwise they wouldn't have had any expectation of what it should or shouldn't do.

- The concept of schema is incompatible with the theories of Bruner (1966) and Vygotsky (1978). Behaviorism would also refute Piaget's schema theory because is cannot be directly observed as it is an internal process. Therefore, they would claim it cannot be objectively measured.
- Piaget studied his own children and the children of his colleagues in Geneva to deduce general principles about the intellectual development of all children. His sample was very small and composed solely of European children from families of high socio-economic status. Researchers have, therefore, questioned the generalisability of his data.
- For Piaget, language is considered secondary to action, i.e., thought precedes language. The Russian psychologist Lev Vygotsky (1978) argues that the development of language and thought go together and that the origin of reasoning has more to do with our ability to communicate with others than with our interaction with the material world.

Piaget's Theory vs Vygotsky

Piaget maintains that cognitive development stems largely from independent explorations in which children construct knowledge of their own.

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Whereas Vygotsky argues that children learn through social interactions, building knowledge by learning from more knowledgeable others such as peers and adults. In other words, Vygotsky believed that culture affects cognitive development.

These factors lead to differences in the education style they recommend: Piaget would argue for the teacher to provide opportunities that challenge the children's existing schemas and for children to be encouraged to discover for themselves.

Alternatively, Vygotsky would recommend that teachers assist the child to progress through the zone of proximal development by using scaffolding.

However, both theories view children as actively constructing their own knowledge of the world; they are not seen as just passively absorbing knowledge.

They also agree that cognitive development involves qualitative changes in thinking, not only a matter of learning more things.

	Piaget	Vygotsky
Sociocultural	Little emphasis	Strong emphasis
Constructivism	Cognitive constructivist	Social constructivist
Stages	Cognitive development follows universal stages	Cognitive development is dependent on social context (no stages)
Learning & Development	The child is a "lone scientist", develops knowledge through own exploration	Learning through social interactions. Child builds knowledge by working with others
Role of Language	Thought drives language development	Language drives cognitive development

	Piaget	Vygotsky
Role of the Teacher	Provide opportunities for children to learn about the world for themselves (discovery learning)	Assist the child to progress through the ZPD by using scaffolding

What is this?

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FAQs

What is cognitive development?

Cognitive development is how a person's ability to think, learn, remember, problem-solve, and make decisions changes over time.

This includes the growth and maturation of the brain, as well as the acquisition and refinement of various mental skills and abilities.

Cognitive development is a major aspect of human development, and both genetic and environmental factors heavily influence it. Key domains of cognitive development include attention, memory, language skills, logical reasoning, and problem-solving.

Various theories, such as those proposed by Jean Piaget and Lev Vygotsky, provide different perspectives on how this complex process unfolds from infancy through adulthood.

What are the 4 stages of Piaget's theory?

Piaget divided children's cognitive development into four stages; each of the stages represents a new way of thinking and understanding the world.

He called them (1) sensorimotor intelligence, (2) preoperational thinking, (3) concrete operational thinking, and (4) formal operational thinking. Each stage is correlated with an age period of childhood, but only approximately.

According to Piaget, intellectual development takes place through stages that occur in a fixed order and which are universal (all children pass through these stages regardless of social or cultural background).

Development can only occur when the brain has matured to a point of "readiness".

What are some of the weaknesses of Piaget's theory?

Cross-cultural studies show that the stages of development (except the formal operational stage) occur in the same order in all cultures suggesting that cognitive development is a product of a biological maturation process.

However, the age at which the stages are reached varies between cultures and individuals, suggesting that social and cultural factors and individual differences influence cognitive development.

What are Piaget's concepts of schemas?

Schemas are mental structures that contain all of the information relating to one aspect of the world around us.

According to Piaget, we are born with a few primitive schemas, such as sucking, which give us the means to interact with the world.

These are physical, but as the child develops, they become mental schemas. These schemas become more complex with experience.

References

Baillargeon, R., & DeVos, J. (1991). Object permanence in young infants: Further evidence. *Child development*, 1227-1246.

What is this?

Report Ad

Bruner, J. S. (1966). *Toward a theory of instruction.* Cambridge, Mass.: Belkapp Press.

Cuban, L., Kirkpatrick, H., & Peck, C. (2001). High access and low use of technologies in high school classrooms: Explaining an apparent paradox. *American Educational Research Journal*, *38*(4), 813-834.

Dasen, P. (1994). Culture and cognitive development from a Piagetian perspective. In W .J. Lonner & R.S. Malpass (Eds.), *Psychology and culture* (pp. 145–149). Boston, MA: Allyn and Bacon.

Gehlbach, H. (2010). The social side of school: Why teachers need social psychology. *Educational Psychology Review*, *22*, 349-362.

What is this?

Report Ad

Hughes, M. (1975). *Egocentrism in preschool children*. Unpublished doctoral dissertation. Edinburgh University.

Inhelder, B., & Piaget, J. (1958). *The growth of logical thinking from childhood to adolescence*. New York: Basic Books.

Keating, D. (1979). Adolescent thinking. In J. Adelson (Ed.), *Handbook of adolescent psychology* (pp. 211-246). New York: Wiley.

Kegan, R. (1982). *The evolving self: Problem and process in human development*. Harvard University Press.

Nielsen. 2014. “Millennials: Technology = Social Connection.” http://www.nielsen.com/content/corporate/us/en/insights/news/2014/millennials-technology-social-connecti on.html.

Passey, D. (2013). *Inclusive technology enhanced learning: Overcoming cognitive, physical, emotional, and geographic challenges*. Routledge.

Piaget, J. (1932). *The moral judgment of the child*. London: Routledge & Kegan Paul.

Piaget, J. (1936). *Origins of intelligence in the child.* London: Routledge & Kegan Paul.

Piaget, J. (1945). *Play, dreams and imitation in childhood*. London: Heinemann.

Piaget, J. (1957). *Construction of reality in the child.* London: Routledge & Kegan Paul.

Piaget, J., & Cook, M. T. (1952). *The origins of intelligence in children*. New York, NY: International University Press.

Piaget, J. (1981). *Intelligence and affectivity: Their relationship during child development.(Trans & Ed TA Brown & CE Kaegi)*. Annual Reviews.

Plowden, B. H. P. (1967). *Children and their primary schools: A report* (Research and Surveys). London, England: HM Stationery Office.

Siegler, R. S., DeLoache, J. S., & Eisenberg, N. (2003). *How children develop* . New York: Worth.

What is this?

Report Ad

Vygotsky, L. S. (1978). *Mind in society: The development of higher psychological processes*. Cambridge, MA: Harvard University Press.

Wadsworth, B. J. (2004). *Piaget's theory of cognitive and affective development: Foundations of constructivism*. New York: Longman.

Further Reading

- BBC Radio Broadcast about the Three Mountains Study
- Piagetian stages: A critical review
- Bronfenbrenner's Ecological Systems Theory

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